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Artem Makoian

Programming: Python 3, C++
Languages: English (C1), Russian (native)
github.com/MakArtKar
scholar.google.com
leetcode.com/MakArtKar

SKILLS

Libraries PyTorch, JAX, Transformers, Hydra, DVC, WandB, OpenCV, Sklearn, Numpy, Pandas, ONNX, Seaborn
Tools SLURM (DDP, FSDP), SQL, Linux, Bash, Git, $\LaTeX$, Markdown

TECHNICAL EXPERIENCE

 **JetBrains** Feb 2025 — Present
ML Researcher / JetBrains Research *with D. Vetrov* Python 3, PyTorch, JAX, transformers, Hydra, Lightning, SLURM

→Masked Diffusion LMs

- Inference-time scaling, diffusion drafters for speculative decoding, curriculum learning (Dream-7B, Qwen3-0.6B MDLM/BD3LM)
- **G-Star**: ICML 2026 main track; multi-node training; PyTorch, JAX

→Latent Diffusion Drafters for speculative decoding: lossless acceleration of LLM inference

- Distillation (flow maps, consistency) and latent space compression for real-time acceleration

 **Yandex** Jun 2023 — Sep 2024
Middle ML Engineer / Geo services Python 3, LangChain, PyTorch, Hydra, OpenCV, YT, Shapely, YQL, Nirvana

→Road segmentation from aerial imagery with IoU 0.93

- Roads post-processing algorithms - reduced number of artifacts from 10% → 0.02% and improved visual quality.
- Safety islands segmentation and vectorization with IoU 0.37

→News classification to create map updates hypotheses. Precision: 0.9, recall: 0.85.

- Developing and maintenance of the news pipeline: site parsing (with LangChain AI agents), structuring data and classifying.

 **Sber Devices** Jul 2022 — Jun 2023
Middle ML Engineer / Computer Vision Platform Python 3, C++, PyTorch, Hydra, DVC, OpenCV, ONNX, torch_pruning

→Background removal for video conferencing with 0.98 Dice score

- Improved human segmentation quality working on segmentation pipeline, pruned model getting 1.5x speed acceleration
- Improved visual quality with post-processing algorithms: aligning the lighting and smoothing out the border.

→Trained a 700 KB model for payment card border segmentation and developed C++ inference pipeline in multiplatform mobile SDK

 **AIRI (Artificial Intelligence Research Institute)** Jul 2022 — Oct 2022
ML Research Intern / GAN adaptation Python 3, PyTorch, Hydra, WandB

- **Few-Shot Image Generation** implementation for few-shot GAN adaptation to a new domain from ~10 images

 **Yandex** Jul 2021 — Oct 2021
ML Researcher / VR Team *with V. Lempitsky* Python 3, PyTorch, Tensorboard

→Trained model that predicts human's alpha mask to improve the quality of a VR scene

- Prepared datasets with human alpha masks, read and compared papers on an alpha matting problem
- Trained an improved **FBA Matting** net without using trimaps

 **Samsung R&D Department** Nov 2020 — Jun 2021
ML Research Intern / Mobile Authentication Python 3, PyTorch, Bash, Numpy, Pandas, Sklearn, Tslearn

→Worked on a pipeline for mobile phone authentication by user's motion patterns with 82.2% ± 9% accuracy - [paper](#).

- Collected, preprocessed and handled data from devices. Used ML and DL models for authentication, got 70% accuracy baseline

EDUCATION

 **PhD in Computer Science (1st year)**, *Constructor University of Bremen, Germany* Sep 2026 — Present

 **Master in Computer Science**, *Constructor University of Bremen, Germany* Sep 2024 — May 2026

 **Bachelor in Computer Science, GPA 86/100**, *Neapolis University of Paphos, Cyprus* Jan 2023 — Jun 2024

Courses: Innovation and Entrepreneurship, Software Project Management, Technical Entrepreneurship, Interest Based Communications, Introduction to Business, Principles of Cybersecurity, Advanced Databases Concepts, Advanced Computer Networks, Distributed Systems

 **Bachelor in Computer Science, GPA 8.7/10**, *Higher School of Economics, Moscow, Russia* Sep 2019 — Jun 2024
Courses: Algebra, Calculus, Algorithms and Data Structures, Probability Theory and Statistics, Discrete Mathematics, Linear Algebra and Geometry, Operating Systems, Logic, Matrix Computations, Optimization in ML, NLP

 **Master's Equivalent**, *Yandex School of Data Analysis, Data Science, Russia* Sep 2020 — May 2022
Courses: Machine Learning, Deep Learning, Deep Vision and Graphics, Computer Vision, 3D Computer Vision, Natural Language Processing, Generative Models, Reinforcement Learning, Bayes Methods in ML, Python, C++

COMPETITIONS

 Candidate for the national team at the **IOI** (International Olympiad in Informatics) May, Nov 2018

 Winner of the national olympiad in **Mathematics** (8th) and **Informatics** (33rd) out of 1M participants Apr 2018, 2019